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DATA SHEET

G2 SETUP
Revision 1.0

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G2 Amiga firmware 1.0 · 2026-11-03

Setup Screen of the G2 Amiga Firmware

A menu-driven setup screen, drawn by the RISC-V firmware on both video heads while the Amiga core is stopped: six pages of items, one for every key of AMIGA.INI, a help pane naming each item's key, pop-ups for lists and numbers, and a pending-change model with Save and reset. Opened with F12, a pad's PS button or the front-panel button.

HIGHLIGHTS

- Six pages: Machine, Drives, Video, Peripherals, Network, System; 48 items.
- Two geometries: 80 × 34 cells at 640 × 480, 40 × 17 cells for 15 kHz monitors.
- Keyboard and DualShock pad control; key repeat 400 ms then 60 ms.
- Pending machine changes in yellow, saved by F10 / Triangle with a confirmation; Discard gives them back.
- Double-buffered on both heads; only changed cells are drawn.

KEY PARAMETERS

PARAMETER	VALUE
Pages	6
Items	48
Wide geometry	80 × 34 cells
Narrow geometry	40 × 17 cells
Font	EGA 8 × 14
Key repeat	400 / 60 ms
Idle refresh	1 s
Settings file	`\amiga\AMIGA.INI`

1 ENTERING AND LEAVING SETUP

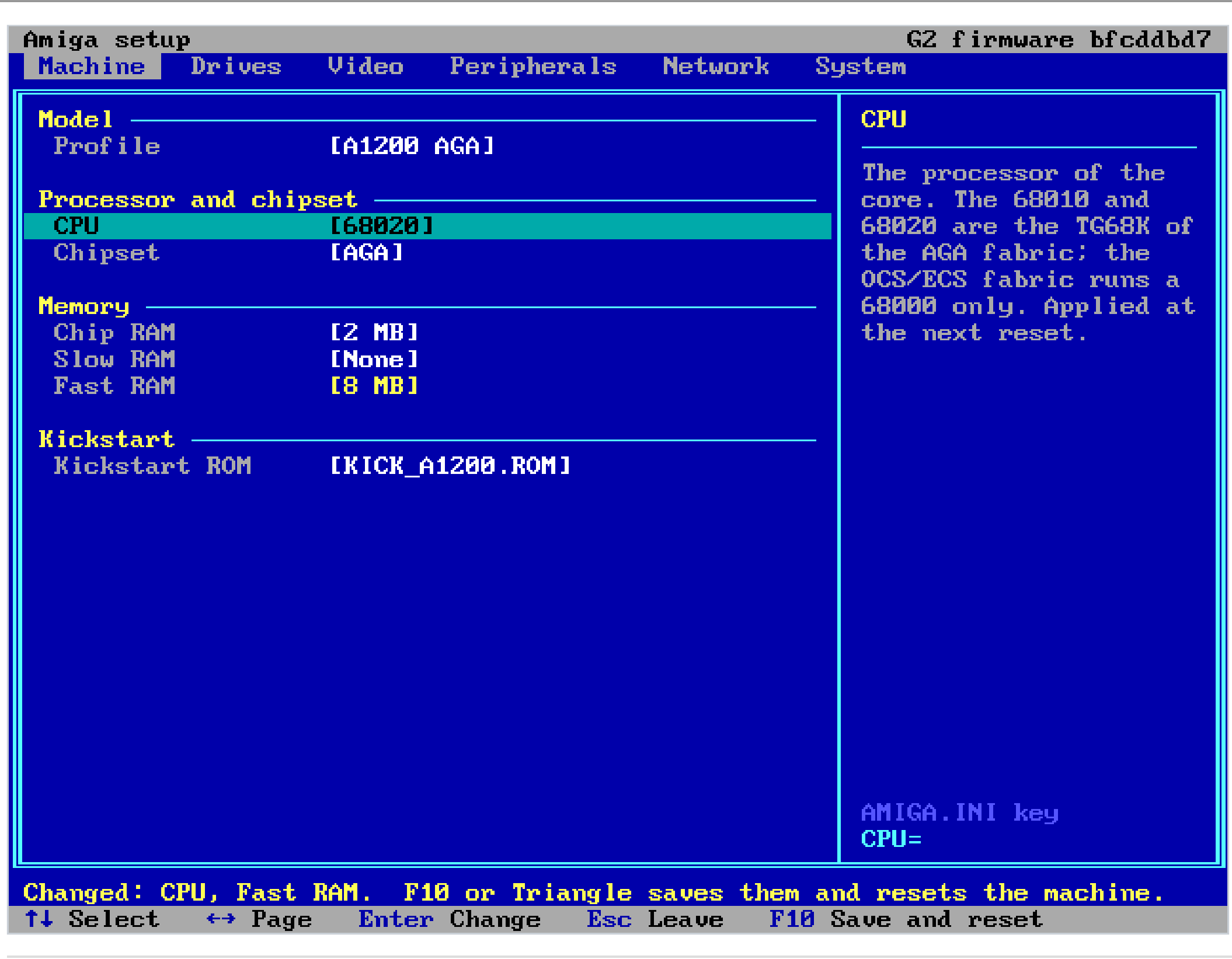
ACTION	KEYBOARD	DUALSHOCK 3 / 4	EFFECT
Open setup	`F12`	PS	The core stops (register CTRL run bit 0) and setup is drawn on both video heads. The unit's front-panel button does the same
Select an item	`↑` `↓`	D-pad, left stick	The selection bar moves to the next item that can be chosen; section titles and information lines are skipped
Change page	`←` `→`	D-pad, left stick	Machine, Drives, Video, Peripherals, Network, System, cyclic
Change the item	`Enter`	Cross, Start	A list of the item's values, a number entry, or the action
Save and reset	`F10`	Triangle	A confirmation dialog, then every pending change is written to AMIGA.INI and the machine starts cold; with nothing pending, the cold start alone
Leave	`Esc`, `F12`	Circle, PS	With nothing pending the machine resumes where it stopped; otherwise a dialog: Save and reset, Discard changes, Back to setup

A held arrow repeats after 400 ms, then every 60 ms. Without a key the screen is drawn again every second, so the clock, the link state and the device lines stay current.

2 LAYOUT

GEOMETRY	CELLS	PIXELS	USED FOR	ARRANGEMENT
Wide	80 × 34	640 × 480 (8 × 14 font)	`VIDEO=SCALE` (default)	Tabs on row 1; items on the left (label 17 cells, value in brackets); help pane on the right (25 cells) with the item's AMIGA.INI key; status line; key bar
Narrow	40 × 17	320 × 238 at 15.7 kHz	`VIDEO=NATIVE`	One page name between ◀ and ▶ with its number; items across the width; help in two rows under them; short key bar

Every string is placed in a field of fixed width and clipped there, never wrapped into the next row; a value that does not fit ends in ». Help text is wrapped at word boundaries inside its pane. Colours: page grey on blue, values white, values that differ from AMIGA.INI yellow, the selection bar black on cyan, pop-ups black on grey with a blue double frame and a black shadow.



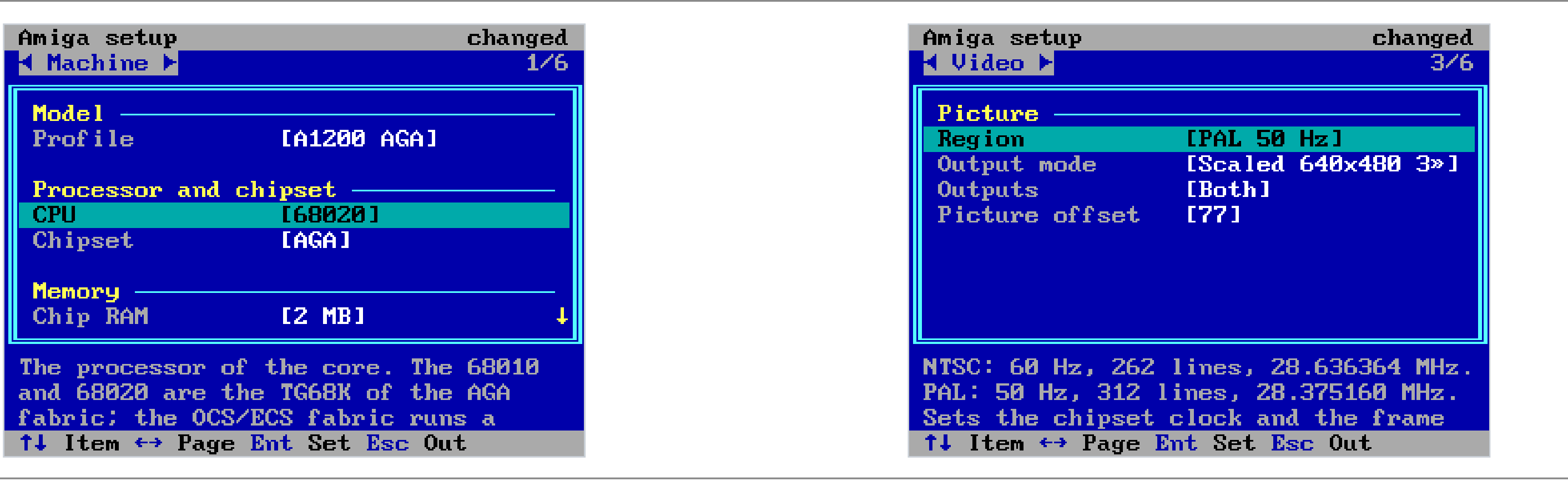
```
Amiga setup                                                    G2 firmware bfcddbd7
 Machine  Drives  Video  Peripherals  Network  System

 Model                                                 CPU
  Profile              [A1200 AGA]
                                                       The processor of the
 Processor and chipset                                 core. The 68010 and
  CPU                  [68020]                         68020 are the TG68K of
  Chipset              [AGA]                           the AGA fabric; the
                                                       OCS/ECS fabric runs a
 Memory                                                68000 only. Applied at
  Chip RAM             [2 MB]                          the next reset.
  Slow RAM             [None]
  Fast RAM             [8 MB]

 Kickstart
  Kickstart ROM        [KICK_A1200.ROM]

                                                       AMIGA.INI key
                                                       CPU=
Changed: CPU, Fast RAM.  F10 or Triangle saves them and resets the machine.
↑↓ Select  ↔ Page  Enter Change  Esc Leave  F10 Save and reset
```

Figure 1. Wide geometry, page Machine. CPU and Fast RAM are pending changes (yellow); the status line names them. Rendered by the firmware's setup.c on a host model of the text console (same font, same cells).



```
Amiga setup                 changed
◀ Machine ▶                     1/6

 Model
 Profile            [A1200 AGA]

 Processor and chipset
 CPU                [68020]
 Chipset            [AGA]

 Memory
 Chip RAM           [2 MB]        ↓
The processor of the core. The 68010
and 68020 are the TG68K of the AGA
fabric; the OCS/ECS fabric runs a
↑↓ Item ↔ Page Ent Set Esc Out
```

```
Amiga setup                 changed
◀ Video ▶                       3/6

 Picture
 Region             [PAL 50 Hz]
 Output mode        [Scaled 640x480 3»]
 Outputs            [Both]
 Picture offset     [77]

NTSC: 60 Hz, 262 lines, 28.636364 MHz.
PAL: 50 Hz, 312 lines, 28.375160 MHz.
Sets the chipset clock and the frame
↑↓ Item ↔ Page Ent Set Esc Out
```

Figure 2. Narrow geometry (40 × 17) for a 15 kHz monitor: pages Machine and Video.

3 PAGES

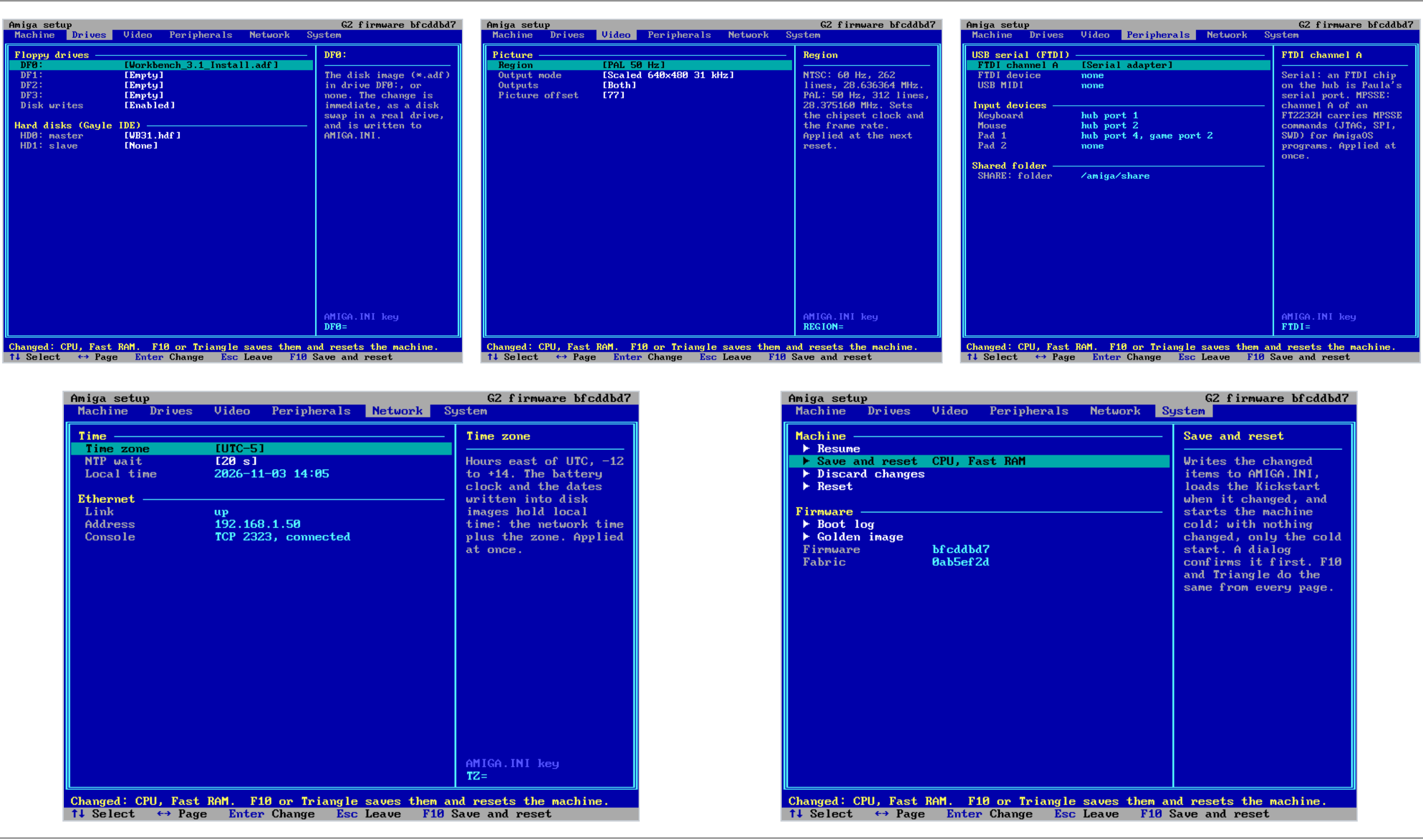


Figure 3. Pages Drives, Video, Peripherals, Network and System (wide geometry). Device lines on Peripherals and Network are filled in from the running firmware.

4 ITEMS

Every key of `\amiga\AMIGA.INI` has an item. Kinds: value (Enter changes it), information (shown only), action (Enter runs it).

4.1 Machine

ITEM	AMIGA.INI	KIND	FUNCTION
Profile	—	value	A set of CPU, chipset, memory and Kickstart: A500 OCS, A500+ ECS, ECS 2 MB, A1200 AGA. Choosing one sets the items below. A profile whose Kickstart image is not in \amiga is refused. Applied at the next reset.
CPU	`CPU=`	value	The processor of the core. The 68010 and 68020 are the TG68K of the AGA fabric; the OCS/ECS fabric runs a 68000 only. Applied at the next reset.
Chipset	`CHIPSET=`	value	OCS as in the A500, OCS as in the A1000, ECS, or AGA. AGA needs the AGA fabric. Applied at the next reset.
Chip RAM	`CHIP=`	value	Memory shared by the CPU and the custom chips: 512 KB, 1 MB, 1.5 MB or 2 MB. Applied at the next reset.
Slow RAM	`SLOW=`	value	Memory at $C00000 on the chip bus, as the A501 trapdoor board: none, 512 KB, 1 MB or 1.5 MB. Applied at the next reset.
Fast RAM	`FAST=`	value	One Zorro II board at $200000: none, 2 MB, 4 MB or 8 MB. Only the CPU reads it. Applied at the next reset.
Kickstart ROM	`KICK=`	value	The ROM image in \amiga (*.rom, 256 KB or 512 KB). Loaded at Save and reset; applied at the next reset.

4.2 Drives

ITEM	AMIGA.INI	KIND	FUNCTION
DF0:	`DF0=`	value	The disk image (*.adf) in drive DF0:, or none. The change is immediate, as a disk swap in a real drive, and is written to AMIGA.INI.
DF1:	`DF1=`	value	The disk image (*.adf) in drive DF1:, or none. The change is immediate, as a disk swap in a real drive, and is written to AMIGA.INI.
DF2:	`DF2=`	value	The disk image (*.adf) in drive DF2:, or none. The change is immediate, as a disk swap in a real drive, and is written to AMIGA.INI.
DF3:	`DF3=`	value	The disk image (*.adf) in drive DF3:, or none. The change is immediate, as a disk swap in a real drive, and is written to AMIGA.INI.
Disk writes	`WRITE=`	value	Enabled: sectors the Amiga writes go to the image files on the stick. Write protected: the floppy drives report a protected disk and image files open read-only. Applied at the next reset.
HD0: master	`HD0=`	value	A hard-disk image (*.hdf with a RigidDiskBlock) on the Gayle IDE port, a new image of 128 MB to 1 GB, or none. The image opens at once and is written to AMIGA.INI; the machine starts cold when setup is left.
HD1: slave	`HD1=`	value	A hard-disk image (*.hdf with a RigidDiskBlock) on the Gayle IDE port, a new image of 128 MB to 1 GB, or none. The image opens at once and is written to AMIGA.INI; the machine starts cold when setup is left.

4.3 Video

ITEM	AMIGA.INI	KIND	FUNCTION
Region	`REGION=`	value	NTSC: 60 Hz, 262 lines, 28.636364 MHz. PAL: 50 Hz, 312 lines, 28.375160 MHz. Sets the chipset clock and the frame rate. Applied at the next reset.
Output mode	`VIDEO=`	value	Scaled: every Amiga mode scaled to 640x480 at 31 kHz. Native: 15 kHz, 640x240 (NTSC) or 640x288 (PAL), for a monitor that takes 15 kHz. Applied at once: setup is shown in the new mode.
Outputs	`SCREEN=`	value	The heads that show the Amiga: the micro-HDMI head, the DVI-I head, or both. Applied when setup is left.
Picture offset	`HOFS=`	value	The first captured sample of a 15 kHz line, in samples of the 14 MHz grid after horizontal blank: 0 to 400. 77 centres the standard display window. Applied when setup is left.

4.4 Peripherals

ITEM	AMIGA.INI	KIND	FUNCTION
FTDI channel A	`FTDI=`	value	Serial: an FTDI chip on the hub is Paula's serial port. MPSSE: channel A of an FT2232H carries MPSSE commands (JTAG, SPI, SWD) for AmigaOS programs. Applied at once.
FTDI device	—	information	The FTDI chip opened and its hub port, or none
USB MIDI	—	information	The class-compliant USB-MIDI interface opened and its hub port, or none
Keyboard	—	information	The hub port of the boot keyboard, or none
Mouse	—	information	The hub port of the boot mouse, or none
Pad 1	—	information	The hub port of the first DualShock pad and the Amiga joystick port it drives (joystick 0, the mouse port)
Pad 2	—	information	The hub port of the second pad (joystick 1, the joystick port), or none
SHARE: folder	`SHARE=`	information	The stick directory shown to AmigaOS as the volume USBStick (AMIGA.INI SHARE=, default /amiga/share)

4.5 Network

ITEM	AMIGA.INI	KIND	FUNCTION
Time zone	`TZ=`	value	Hours east of UTC, -12 to +14. The battery clock and the dates written into disk images hold local time: the network time plus the zone. Applied at once.
NTP wait	`NTPWAIT=`	value	Seconds after power-up the first start of the machine waits for network time, so Kickstart finds the clock set; 0 starts at once. Applied at the next power-up.
Local time	—	information	Date and time from the network plus the zone, or not set
Link	—	information	Ethernet link: up with its speed, or down
Address	—	information	The IPv4 address from DHCP
Console	—	information	The network console (TCP 2323): waiting or connected

4.6 System

ITEM	AMIGA.INI	KIND	FUNCTION
Resume	—	action	The machine continues where it stopped. With changes not saved, a dialog offers to save them and reset, or to discard them.
Save and reset	—	action	Writes the changed items to AMIGA.INI, loads the Kickstart when it changed, and starts the machine cold; with nothing changed, only the cold start. A dialog confirms it first. F10 and Triangle do the same from every page.
Discard changes	—	action	Gives back every item changed since setup was opened, and resumes.
Reset	—	action	Starts the machine cold with the saved settings; changes not saved are discarded.
Boot log	—	action	The console lines of this boot, as \amiga\BOOT.LOG holds them.
Golden image	—	action	Closes every image file and returns the unit to the golden image in flash.
Firmware	—	information	The ID of the running AMIGA.FW (the first 8 hex digits of its source commit)
Fabric	—	information	The build ID of the FPGA fabric

5 POP-UPS

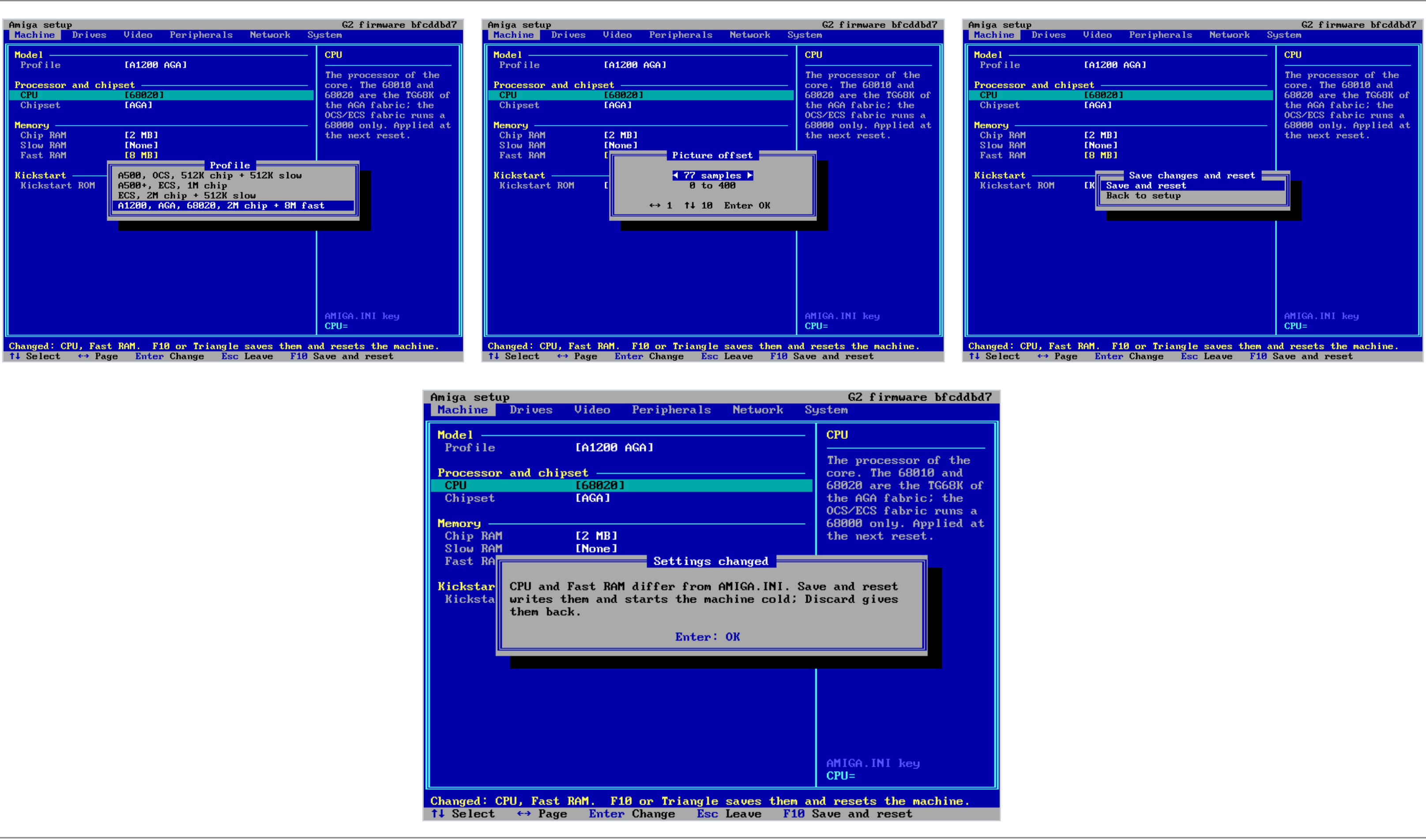


Figure 4. A list (Profile), a number (Picture offset, ←→ by 1, ↑↓ by 10), the F10 dialog and a message. Escape closes a pop-up without a change.

POP-UP	KEYS	RESULT
List	↑ ↓ select, Enter choose, Esc cancel	The chosen value; the list scrolls with arrows in the frame when it is longer than the screen
Number	← → ±1, ↑ ↓ ±step, Enter accept, Esc cancel	A value inside the item's range
Message	Enter or Esc	Closes the message
Save and reset (F10)	Enter on Save and reset, or Back to setup	Pending changes written and a cold start, or nothing

6 WHEN A CHANGE TAKES EFFECT

GROUP	ITEMS	WRITTEN TO AMIGA.INI	TAKES EFFECT
Machine configuration	Profile, CPU, Chipset, Chip, Slow and Fast RAM, Kickstart, Region, Disk writes	At Save and reset	At the cold start that follows; Discard gives the values back
Drives	DF0:–DF3:, HD0:, HD1:	At once	Floppies at once (a disk change); a hard-disk change forces a cold start when setup is left
Picture	Output mode, Outputs, Picture offset	At once	Output mode at once; outputs and offset when setup is left
Other	FTDI channel A, Time zone, NTP wait	At once	At once; NTP wait at the next power-up

7 DRAWING

PROPERTY	IMPLEMENTATION
Buffers	Two text frame buffers on each video head (DDR2 chip B), shown alternately; the swap waits for the vertical blank on both heads
Composition	Every screen is one composed frame: the cells change in memory, and only at the end of the frame are the glyphs drawn
Changed cells	Each buffer keeps a shadow of the characters and colours it shows; a frame draws only the cells that differ from the back buffer's shadow. Moving the selection bar redraws two rows
Font	IBM EGA 8 × 14, code page 437 line-drawing characters for frames and arrows

8 LIMITS AND UNTESTED ITEMS

ITEM	STATE
Wide geometry, every page and pop-up, F10 dialog	Run on a G2 revision C unit, firmware 0x92A8A0FD
Narrow geometry (VIDEO=NATIVE)	Rendered on the host model only; untested on hardware (needs a 15 kHz monitor)
Frame time of a redraw	Not measured
Keyboards	USB boot-protocol keyboards; see the input data sheet for the speeds the USB host serves

Item table and help text from the firmware source (fw/amiga/setup_items.h); figures drawn by fw/amiga/setup.c on a host model of the text console. Repository mikewolak/pano-g2-amiga, release 1.0.